

Customer Shopping Behavior Analysis

Turning 3,900 purchases into clear, actionable business strategy.

Uncovering patterns in customer behavior to identify opportunities, refine strategy, and support confident business decisions.

PYTHON

MYSQL

POWER BI

Dataset Overview

3,900

Total Purchases

Comprehensive transaction data

18

Data Columns

Customer demographics & behavior

37

Missing Values

Only in Review Rating column

Data Preparation Process

01

Data Loading

Imported dataset using pandas

02

Initial Exploration

Structure check and summary statistics

03

Missing Data Handling

Imputed Review Rating with median values

04

Feature Engineering

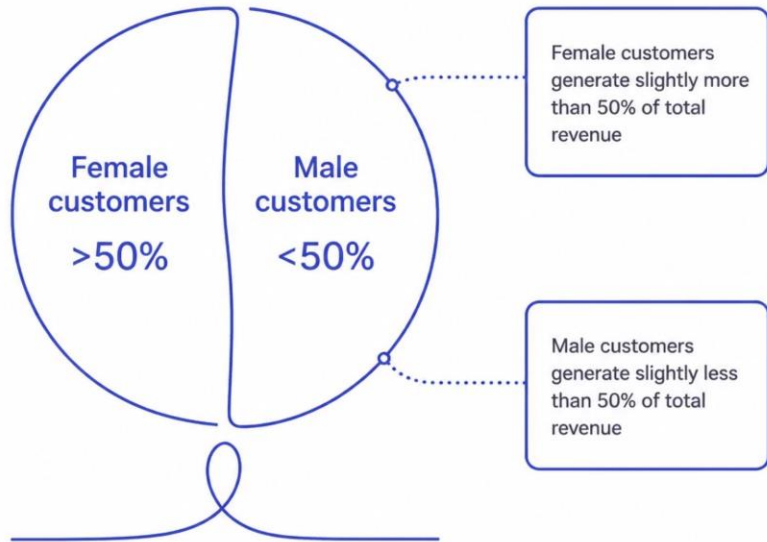
Created age groups and purchase frequency

05

Database Integration

Connected to MySQL for analysis

Revenue by Gender



Key Insight

Female customers generate **slightly higher total revenue** than male customers.

Gender-based marketing strategies could optimize revenue streams.

High-Value Discount Customers @ Top-Rated Products

High-Value Discount CustomersTop-Rated Products

Above-average spenders who use discounts
represent high-value, price-conscious customers.
Target them with personalized offers.

- **Blouse** – Highest customer satisfaction (5★)
- **Dress** – Consistently excellent reviews (5★)
- **Shirt** – Strong customer approval (4★)

Shipping & Subscription Impact

Express Shipping

Avg. purchase: **\$65**

vs. Standard: \$58

Express customers spend **12% more** per transaction.

Subscriber Spend

Subscribers spend **68% higher** than non-subscribers.

Revenue Share

Subscription customers contribute **45%** of total revenue.

Loyalty Rate

78% repeat purchase frequency among subscribers.

Customer Segmentation



Segment Breakdown

- **Loyal (15%)** – High-value customers
- **Returning (35%)** – Regular shoppers
- **New (50%)** – First-time buyers

Focus on converting New → Returning → Loyal to maximize lifetime value.

Interactive Power BI Dashboard



Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits for subscribers



Loyalty Programs

Reward repeat buyers to increase retention



Targeted Marketing

Focus on high-revenue segments and express users



Product Positioning

Highlight top-rated products in campaigns